2022 International Conference on Cloud Computing, Big Data Applications and Software Engineering (CBASE) | 979-8-3503-9639-3/22/$31.00 ©2022 IEEE | DOI: 10.1109/CBASE57816.2022.00045

Bayesian optimization for robust engineering design with noise factors

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Abstract. **Robust parametric engineering design with noise is critical due to the biasing effect that noise factors can have on the system during the actual process of engineering design. Most Bayesian algorithms for robust parameter design use a quadratic loss function as a measure of quality characteristics, which is not always realistic. In this paper, two Bayesian optimization algorithms are developed for general loss functions for the first time, and Monte Carlo sampling is used to implement the computation of the proposed acquisition functions for the optimization process. The effectiveness of the two proposed algorithms is demonstrated in three arithmetic cases.**

Keywords: Bayesian optimization; robust parameter design; general loss function

I. Introduction

In engineering design, there are some uncontrollable and changeable factors that can produce non-negligible disturbance to the system, which are called noise factors. Therefore, we classify the design variables into control and noise variables. Control variables are usually determined by the product designer and generally refer to the design nominal value of a particular product. Noise variables are divided into internal noise and external noise, which originate from deviations between actual operation and nominal values, or changes in the external environment respectively. It is critical to consider noise factors in actual engineering design.

Robust parameter design (RPD) is a method to improve the robustness of a product by optimizing design variables, which is to find an optimal set of design parameters to reduce fluctuations in quality characteristics. The objective of RPD is to minimize the expected quality loss (EQL), which is the average quality loss with respect to the change of noise factors. RPD originally came from the key technology in Taguchi's design method, and then many experts and scholars began to expand it innovatively.

The Bayesian optimization algorithm starts with the establishment of a probabilistic surrogate model, based on the distribution between the objective and the design parameters. Sampling is then performed according to the established model, and a criterion needs to be followed to determine the updated sampling points by maximizing the value of the acquisition function (AF). Sampling points are then added to the original model to update the model. As the number of trials increases, the prior probability of the objective function is continuously corrected so that the proxy model representing the unknown function is more accurate and the optimal solution can be finally found. Therefore, the Bayesian optimization algorithm consists of two parts, establishing the probabilistic surrogate model and developing the AF. The surrogate model used in this paper is the Kriging model (called Gaussian process model in machine learning), and the two AFs are the core of the algorithms we proposed. The two improved algorithms proposed in this paper are based on expected improvement (EI) and upper-confidence bound (UCB), which can perform global optimization to avoid falling into local optimum because of both exploratory and exploitation properties. Jones and Schonlau [7] proposed the EI in 1998, also known as the efficient global optimization algorithm. Emmerich et al.[3], Keane[8], Knowles[9], Shinkyu Jeong & Obayashi [13] extend the EI criterion to multi-objective optimization problems. Auer and Cesa-Bianchi [2] first proposed the UCB machine learning algorithm, which has been widely popularized. Srinivas et al. [15] proposed the UCB algorithm for Gaussian process optimization (GP-UCB) and compared it with other heuristics based on the Gaussian process model to demonstrate the superiority of the algorithm.

Most of the studies in RPD assume the loss function as a quadratic function. Ames et al.[1], Fallah Nezhad & Farsani [4], Jenoui & Abouabdellah [6] employ quadratic loss function for RPD research. As for the research of Bayesian optimization, Frazier [5], Shahriari et al.[12], Zhan & Xing [19] conducted a comprehensive investigation on AFs of various engineering optimization problems. The quadratic loss function does not always approximate the true loss function. Since the quadratic loss is unbounded, for some assumed noise factor distribution, the expected loss does not exist. The loss function being constructed can only better approximate the true loss function if it is flexible and suitable for the particular problem, since using different loss functions will produce different expected loss values and optimal settings. Therefore, it is necessary to construct a general loss function. Spiring & Yeung [14], Tan [16] proposed bounded general loss functions to compute the expected quality loss. There is no literature on constructing general loss functions for RPD problems for Bayesian optimization of quality design. To fill this gap, this paper proposes the Bayesian optimization methods for quality design developed for general loss functions. We propose two different algorithms, the EI algorithm for general loss (EI-GL) and the GP-UCB algorithm based on general loss (UCB-GL), respectively. And three numerical arithmetic tests were done to demonstrate the effectiveness of both algorithms.

II. Methodology

In this section, we first introduce the expressions of general bounded loss functions that can be employed to measure the performance of quality, as described in subsection A. We also propose two fast computational methods for Bayesian

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optimization to achieve an optimal design of the bounded loss function, in other words, two different AFs are developed to find the update point. The algorithms are both performed in two steps, first finding the control factor and then the noise factor, which are shown in Subsection B and Subsection C, respectively. We give a summary in subsection D.

A. General loss function

The quality loss function studied in this paper belongs to the nominal-the-best quality characteristic, which means that the closer to the target design the smaller the quality loss. According to Tan [17], the general formula for the quality loss function is shown below:

$$L(y) = c\big(M - g(y)\big) \tag{1}$$

Where y is the output, g is a PDF, c is a constant and M is the maximum value of g. There are many selections about the PDF of g, such as normal function, gamma function, beta function, Tukey function, etc. The correct selection of the appropriate density function plays a crucial role in the robust parameter design problem. In this paper, we use the gamma function and the beta function as the density function.

B. Seeking for control factors

This section introduces EI-GL and UCB-GL algorithms, which gives two different AFs by Monte Carlo sampling to find the location of the control variables.

1) EI algorithm for general loss (EI-GL)

In this section, we propose the EI algorithm for general bounded loss functions to find the optimal position of the control variables by maximizing the AF as follows:

$$\boldsymbol{x}_c^* = argmax\ EIL(\boldsymbol{x}_c) \tag{2}$$

Tan [17] mentions that the EI for the quality loss function can be extended according to the ordinary EI expression as follows:

$$EIL(\boldsymbol{x}_c) = E\left[\big(\tau - L(\boldsymbol{x}_c)\big)^+ \middle| \boldsymbol{Y}(D_n)\right] \tag{3}$$

Where $\tau = \min\{L^n(\boldsymbol{x}^1), \dots, L^n(\boldsymbol{x}^n)\}$, $L^n(x)$ represents the posterior mean of L. D_n represents the initial design with n points $\{\boldsymbol{x}^1, .., \boldsymbol{x}^n\}$. For equation (3), the calculation steps are as follows. In step1, the control variable x_c is discretized into N design points in the range area. For each control variable point x_c, we predict the mean vector $\boldsymbol{\mu}_y$ and variance matrix $\boldsymbol{\Sigma}_y$ for that point based on the constructed GP model in step2. In step3, we use Monte Carlo sampling to generate n sample paths of output y, where the sample paths follow multivariate normal distribution, and the mean and variance matrices are $\boldsymbol{\mu}_y$ and $\boldsymbol{\Sigma}_y$. According to the formula (1) of the quality loss function of the nominal-the-best, in step4 we choose the appropriate function such as beta function, gamma function that can calculate the value of n L. In step 5, we calculate the mean value of $(\tau - L)^+$. EIL value from equation (3) would be obtained and we finally find the optimal $\boldsymbol{x}_c^*$ by equation (2).

2) UCB algorithm for general loss (UCB-GL)

We extend the traditional GP-UCB algorithm based on its problem of maximizing the output y for the ordinary problem of minimizing the quality loss L to be used to find the optimal control factor x_c. Our proposed optimization criterion is shown in the following equation:

$$\boldsymbol{x}_c^* = argmax\ UCB(\boldsymbol{x}_c) \tag{4}$$

Where the calculation of UCB can be expressed by

$$UCB(\boldsymbol{x}_c) = \left[-\mu_L(\boldsymbol{x}_c) + \sqrt{\beta_k}\,\sigma_L(\boldsymbol{x}_c)\right] \tag{5}$$

According to Srinivas et al. [15] , $\beta_k = 2\log\left(\frac{|D|k^2\pi^2}{6\delta}\right)$, k is the number of iteration. Similar to the EI-GL algorithm, the calculation of u_L and σ_L is also derived using Monte Carlo sampling. The specific steps 1-4 are the same as steps 1-4 in part 2.2.1. After deriving the values of n Ls, the mean μ_L and variance σ_L^2 of these n Ls are calculated in step 5, and then the UCB values of N points can be obtained by calculating Equation (5). The last step selects the $\boldsymbol{x}_c^*$ corresponding to the largest value of UCB among these N control variable points as the optimal control variable design. It can be seen from Eq. (5) that the criterion prefers a smaller mean value of L and a larger standard deviation, which is satisfying both exploitation and exploration characteristics.

C. Seeking for noise factors

After finding the control factors, we propose a random sampling approach to find the noise factors. We use the randi function in MATLAB to generate pseudo-random integers uniformly distributed within n_{x_e}, where n_{x_e} is the dimension of the vector x_e. This random integer is then used as an index of vectors x_e to find the value of the corresponding noise factor.

D. Summary for two algorithms

In this section, the specific steps of the EI-GL and UCB-GL algorithms are summarized as shown in Figure 1. We assume that the total number of iterations is N. In step1, we identify the ranges region of internal noise factors and their probability distributions. We determine Gaussian quadrature points and weights in step2. A GP model is constructed with initial design from Latin hypercube design (LHD) and corresponding response in step3. We estimate posterior mean and variance of response $Y(\boldsymbol{x}_c)$ in step4. We seek the control factors and noise factors are in step5 and step6. We determine the optimal control factors by maximizing EIL($\boldsymbol{x}_c$) or UCB($\boldsymbol{x}_c$), and determine the noise factors by random sampling. After completing step6, we get $\boldsymbol{x}_{add} = \boldsymbol{x}_c^{n+k} + \boldsymbol{x}_e^{n+k}, k = 1, \dots, N$. In step7, we run the computer model at x_{add} to obtain $Y(\boldsymbol{x}_{add})$. We update the GP model in step8. In step9, we compare the number of iterations with N, if it smaller, these two algorithms would go back to step4 to complete the new iterations. Otherwise, we could estimate the optimal design for RPD utilizing the final model in step10.

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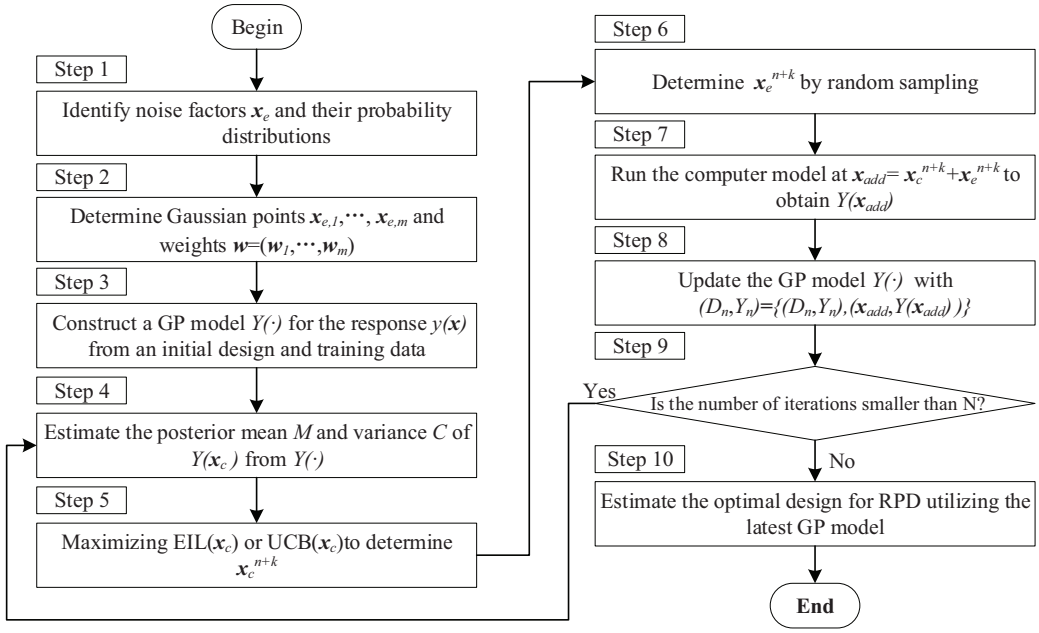

Fig. 1 Flowchart of the EI-GL and UCB-CL algorithms for RPD

III. Experiment analysis

In this section, we test the performance of three different numerical functions in the proposed two Bayesian optimization algorithms by comparing them with one-shot Kriging algorithm. In these three different numerical examples, we use GAP as a measure of the algorithm, as shown in the following equation:

$$GAP = EQL(x_c^*) - \min\{EQL(x_c), x_c \in \chi\} \tag{6}$$

$$x_c^* = argmin\{L^n(x_c), x_c \epsilon \chi\} \tag{7}$$

$$EQL(x_c) = E_{x_e}\big(L(y)\big) \tag{8}$$

Where EQL is the expected value of quality loss and the optimal control factor x_c^* is estimated by minimizing of the posterior loss mean value over the experiment domain χ. GAP as a measure of the performance of the three algorithms quantifies the distance of the minimum EQL value between the estimated results by the model we constructed and the true minimum EQL value. Our optimized model is more accurate as GAP approaches 0, which means the better when the value of GAP is smaller.

A. Example1: Rosenbrock function

In this section, we test the performance of the two proposed Bayesian optimization algorithms of EI-GL and UCB-GL, on a two-dimensional Rosenbrock function [10]. We first normalize the input on [0,1], and we obtain the following test function.

$$f(x_1,x_2) = 100\big((4x_2-2)-(4x_1-2)^2\big)^2 + (4x_1-2)^2, 0 < x_1, x_2 < 1. \tag{9}$$

We consider the internal noise, which means that the input variables x_1 and x_2 are both the sum of the control variable x_c and the noise variable x_e, $x_i = x_{c,i} + x_{e,i}, i = 1,2$. And we assume that $x_e s$ both follow a uniform distribution on [0,0.3], then we can conclude that the range of x_c is [0,0.7]. Latin hypercube sampling was used to obtain our initial experimental design points, and we repeated the experiment ten times to eliminate the effect of randomness with giving ten different LHDs. we set the number of initial points as 3 and the maximum number of iterations as 10. 5-point Gaussian quadrature is constructed for noise factors to approximate response. We use a gamma function with parameters k=9 and theta=1 as PDF g, transform the output y linearly on [0, 20], and let c=1.

In order to visually analyze the superiority of the two proposed algorithms, the results of the experiments repeated ten times are plotted in figure 2, which reflects the trend of the mean value of GAP with the number of iterations represented by k. We use the curve with the green cross symbol to represent the one-shot Kriging algorithm, the purple circle to represent the EI-GL algorithm, and the red triangle to represent the UCB-GL algorithm. From the figure 2, it is seen that as the number of iterations increases, all three curves show an overall decreasing trend and finally converge to 0. Due to the characteristic that the smaller the indicator GAP the more accurate the model is, we can conclude that it is getting closer to find the optimal design as the number of experimental sample points increases. Comparing these three algorithms, the green curve is always above the other two curves, which reflects that our proposed two algorithms converge faster and perform better than the one-shot Kriging algorithm. The purple curve converges in the sixth iteration, while the red and green GAP

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approaches 0 in the eighth iteration. The red curve is only below the purple curve at the first and third iterations, and below the purple curve at the rest of the iterations. In summary, both of the proposed algorithms show their better performance in this example. EI-GL algorithm has the best overall performance, followed by UCB-GL algorithm and one-shot Kriging algorithm.

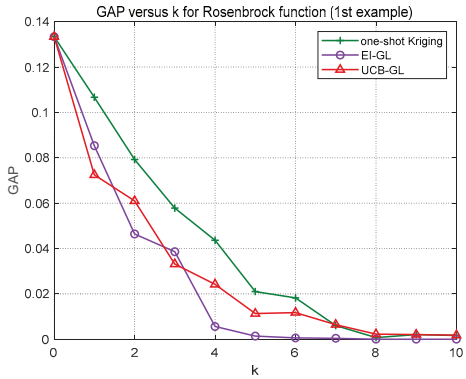

Fig.2 Averages of GAP versus k of 3 algorithms for the Rosenbrock function

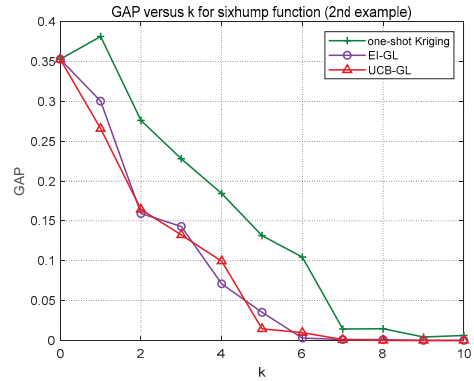

Fig.3 Averages of GAP versus k of 3 algorithms for the Six-hump camel-back function

B. *Example2: Six-hump camel-back function*

In this part, we test the performance of the algorithm on a two-dimensional six-hump Camel-back function [18], and we change the range of inputs to$[0,1]^2$, which is given by:

$$f(x_1, x_2) = 4x_1^2 - \frac{21}{10}x_1^4 + \frac{1}{3}x_1^6 + x_1x_2 - 4x_2^4, 0 < x_1, x_2 < 1. \quad (10)$$

Also consider the internal noise of the two output variables and set the noise variables to follow a uniform distribution on [-0.2, 0.2], then the range of x_c can be obtained as [0.2, 0.8]. Repeat the experiment 10 times, the size of the LHD is 3 and the maximum iteration number is 10. 5-point Gaussian quadrature is constructed for noise factors to approximate response. Let c=1 and normalize the function output y to [0,1], and use the beta function with $\alpha = 2, \beta = 5$ as PDF g.

In Figure 3, we compare the performance results of the three algorithms on six-hump camel-back function. All three algorithms show a downward trend, which also verifies that the model is more accurate with the increase of iterations in this example. The green curve is always above the curves of the other two algorithms, and in the tenth iteration it is very close to the horizontal axis but still does not converge to 0. The purple curve converges in the sixth iteration, and the red curve converges in the seventh iteration, and the red curve and the purple curve have relatively similar descent paths. It can be concluded that the two Bayesian optimization algorithms proposed by us have better convergence effects in this example compared with one-shot Kriging algorithm.

C. *Example3: Sasena function*

In this section, we tested the performance of the three algorithms through a two-dimensional Sasena function [11]. We first normalize 2-D inputs into $[0,1]^2$, and the normalized function is given by

$$f(x_1, x_2) = 2 + 0.01(5x_2 - 25x_1^2)^2 + (1 - 5x_1) + 2(2 - 5x_2)^2 + 7\sin(2.5x_1)\sin(17.5x_1x_2), \quad 0 < x_1, x_2 < 1. \quad (11)$$

For two inputs, we assume that they both have internal noise x_e, and follow a uniform distribution over [-0.2,0.2]. Therefore, we get the ranges of control factors for [0.2,0.8]. The size of initial design from LHDs set is 3 and the maximum number of iterations is 10. 5-point Gaussian quadrature is constructed for noise factors to approximate response. We also test ten different LHDs to compare 3 algorithms to remove the randomness of initial design. As the same as example 2, we let c=1 and normalize the function output y to [0,1], and use the beta function with $\alpha = 2, \beta = 5$ as PDF g.

The mean GAP results of 10 tests of Sasena function are sorted out as shown in Figure 4. The curves of the three algorithms gradually decline. The mean GAP value of the green curve is larger than that of the other two algorithms in each iteration, and it has not yet converged in the 10th iteration, while the purple and red curves converge in the sixth iteration. Compared with one-shot Kriging, the two proposed algorithms have significantly faster convergence speed. The performance of EI-GL and UCB-GL curves on this algorithm is similar, while UCB-GL is slightly better.

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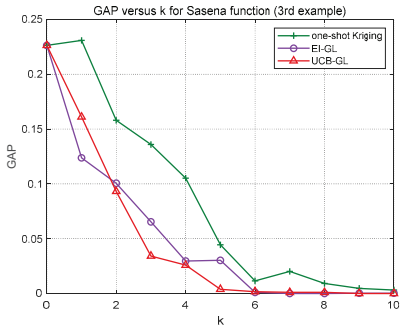

Fig.4 Averages of GAP versus k of 3 algorithms for the Sasena function

IV. Conclusions

In this paper, we propose two Bayesian optimization algorithms for solving robust engineering quality design problems, which develop two Afs for general loss functions. Approximate calculation of AFs using Monte Carlo sampling method to find the control factor. The random sampling method is used to find the noise factor. Implementing sequential addition of trial sampling points to update the model and finally find the optimal design.

We tested the two proposed algorithms EI-GL and UCB-GL with three different two-dimensional arithmetic examples. The three cases use different gamma functions and beta functions as loss functions to measure the quality characteristics and compare with one-shot Kriging. All three examples demonstrate that both of our proposed algorithms have faster convergence, which means that fewer resources can be used to find the optimal solution.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 71902089) and the Natural Science Foundation of Jiangsu Province (No. BK20190389).

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